

Testing Report

Human Development Index (HDI) Prediction Using Machine Learning

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1. Introduction

Testing is an important phase of software development that verifies whether the application works correctly and satisfies the project requirements. It helps identify errors, improve software quality, and ensure reliable performance before deployment.

For the Human Development Index (HDI) Prediction project, testing was carried out on the Machine Learning model, Flask web application, user interface, and prediction functionality to ensure accurate and consistent results.

2. Objectives of Testing

The main objectives of testing are:

- Verify that the application functions correctly.
 - Ensure accurate HDI prediction.
 - Validate user input.
 - Detect errors and bugs.
 - Improve reliability and usability.
 - Confirm that all project requirements are satisfied.
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3. Types of Testing Performed

The following testing methods were applied during the project.

Testing Type	Purpose
Unit Testing	Test individual modules
Functional Testing	Verify application features
Input Validation Testing	Check invalid and empty inputs
Integration Testing	Verify interaction between Flask and ML model
User Interface Testing	Ensure responsive design

Testing Type	Purpose
Performance Testing	Check response time
System Testing	Verify complete application workflow

4. Test Environment

Component	Details
Operating System	Windows 11
Programming Language	Python 3.10+
IDE	Visual Studio Code
Framework	Flask
Browser	Google Chrome
Machine Learning Library	Scikit-learn
Dataset	HDI Dataset

5. Functional Test Cases

Test Case ID	Test Description	Expected Result	Actual Result	Status
TC-01	Launch Flask Application	Home page opens successfully	Successfully opened	✓ Pass
TC-02	Enter valid input values	Prediction generated	Prediction displayed	✓ Pass
TC-03	Empty input fields	Error message displayed	Correct message shown	✓ Pass
TC-04	Invalid numeric input	Validation error	Validation successful	✓ Pass
TC-05	Large GNI value	Prediction generated	Successful	✓ Pass
TC-06	Refresh page	Application reloads correctly	Working correctly	✓ Pass

6. Input Validation Testing

The application validates user inputs before sending them to the prediction model.

Validation checks include:

- Empty fields
- Negative values
- Invalid numbers
- Missing input values
- Incorrect data types

Proper error messages are displayed whenever invalid input is detected.

7. User Interface Testing

The interface was tested for usability and responsiveness.

The following features were verified:

- Responsive layout
- Mobile-friendly design
- Button functionality
- Input fields
- Prediction result display
- Color consistency
- Typography
- Navigation

All UI components worked as expected.

8. Machine Learning Model Testing

The trained Linear Regression model was evaluated using standard regression metrics.

Metric	Value
Algorithm	Linear Regression
R ² Score	0.9892
Prediction Accuracy	Excellent
Training Status	Successful

The model produced consistent predictions for different test inputs.

9. Sample Test Execution

Input

Parameter	Value
Country	India
Life Expectancy	70
Mean Years of Schooling	7
Expected Years of Schooling	13
GNI per Capita	7000

Output

Result	Value
Predicted HDI	0.68 (Approx.)
Category	Medium Human Development

The prediction matched the expected behavior of the trained model.

10. Bug Summary

During development, a few minor issues were identified and resolved.

Issue	Solution
Invalid input handling	Added input validation
UI alignment	Improved CSS styling
Prediction formatting	Rounded output values
Responsive layout	Updated media queries

No critical issues remained after testing.

11. Testing Outcome

The testing process confirmed that:

- The Flask application launches successfully.
- User inputs are validated correctly.
- The machine learning model generates accurate predictions.
- The user interface is responsive and easy to use.
- The prediction results are displayed correctly.
- The complete workflow operates without major errors.

12. Conclusion

Testing played a crucial role in ensuring the quality and reliability of the Human Development Index Prediction System. Various testing techniques were performed to verify the correctness of the machine learning model, Flask application, and user interface.

The successful completion of all test cases demonstrates that the application functions as intended and is suitable for academic submission, GitHub portfolio presentation, and future enhancements.